

Report
Of
AI Auto-Apply Job Applicator & Career Coach

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Batch: 4CSE5

2026-27

CANDIDATE'S DECLARATION

I, Nishant Tomar do hereby declare that the internship report titled “**AI Auto-Apply Job Applicator & Career Coach**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name – Nishant Tomar

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

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INTERNSHIP COMPLETION CERTIFICATE



Cognifyz IT Solutions

Internship Completion Certificate

Date -06/07/2026

This is to certify that **Nishant Tomar (Intern ID: CTI/AI/C367943)** currently pursuing B.Tech from The IILM University, was working as a **Machine Learning Intern** with Cognifyz IT Solutions Pvt. Ltd. from 06/06/2026 to 06/07/2026.

During this period, he has served as a Machine Learning Intern and has displayed remarkable dedication, sincerity, and a strong desire to learn. He has exhibited exceptional coordination skills and effective communication abilities. Moreover, his attention to detail has been truly impressive.

He has consistently approached new assignments and challenges with enthusiasm, showcasing His passion for Machine Learning. His commitment and willingness to acquire new knowledge and skills have been evident throughout His internship.

We extend our best wishes to Nishant Tomar for a successful future, and we have no doubt that he will continue to excel in the field of Machine Learning.


With Regards,
Cognifyz IT Solutions Pvt. Ltd.



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4.1 Introduction

The rapid advancement of technology has significantly transformed the job recruitment process. However, the process of applying for jobs remains repetitive, time-consuming, and inefficient for many candidates. Job seekers often need to manually search for opportunities, tailor resumes, and repeatedly fill out application forms across multiple platforms. In addition, candidates frequently lack personalized guidance for improving their resumes and preparing for interviews.

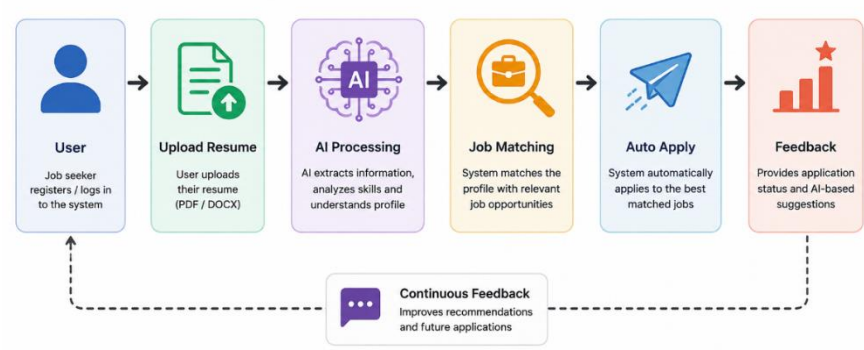
To address these challenges, the project titled “**AI Auto-Apply Job Applicator & Career Coach**” has been developed. This system aims to automate the job application process using artificial intelligence while also providing intelligent career guidance and interview preparation tools. The platform integrates multiple functionalities, including automated job applications, resume analysis, AI-based career suggestions, and voice-enabled mock interviews.

The system is built using a modern full-stack architecture, where the frontend is developed using Next.js and the backend is powered by Spring Boot. Artificial intelligence capabilities are integrated using the Google Gemini API, enabling dynamic job matching, resume evaluation, and conversational assistance. Additionally, the system incorporates speech recognition and synthesis technologies to simulate real interview environments.

One of the key features of this project is its ability to handle API limitations through fallback mechanisms, ensuring uninterrupted functionality even under constrained conditions. The platform also supports multiple resumes tailored for different job roles, enhancing the effectiveness of job applications.

Overall, this project provides a comprehensive solution for job seekers by reducing manual effort, improving application efficiency, and enhancing interview readiness through intelligent automation and AI-driven insights.

Figure 4.1: Overview of AI Auto-Apply System



4.2 Organization Profile

The internship was completed at **Cognifyz IT Solutions Pvt. Ltd.**, a technology-driven organization focused on delivering innovative software solutions. The company specializes in leveraging modern technologies such as Artificial Intelligence, Machine Learning, and Web Development to solve real-world problems and enhance user productivity.

Cognifyz IT Solutions operates with a strong emphasis on data-driven intelligence, as reflected in its vision of integrating advanced analytics and AI into practical applications. The organization is committed to building scalable, efficient, and user-centric systems that cater to evolving industry needs.

The work environment at Cognifyz is collaborative and learning-oriented. Interns are provided with opportunities to work on real-time projects, enabling them to gain hands-on experience in software development and AI-based solutions. The organization encourages innovation, critical thinking, and continuous skill development through mentorship and guided project work.

During the internship period, the role involved working as a **Machine Learning Intern**, contributing to the development of intelligent systems and gaining exposure to full-stack technologies. The internship project titled “**AI Auto-Apply Job Applicator & Career Coach**” aligns with the organization’s objective of using Artificial Intelligence to automate complex processes and improve efficiency.

Overall, Cognifyz IT Solutions provides a strong platform for aspiring developers to enhance their technical skills, understand industry practices, and gain practical experience in building modern software applications.

Table 4.1: Organization Profile

Particulars	Details
Organization Name	Cognifyz IT Solutions Pvt. Ltd.
Organization Type	Software Development / IT Solutions
Domain	Artificial Intelligence, Machine Learning
Nature of Work	Development of intelligent and data-driven applications
Technologies Used	AI/ML, Next.js, React, Spring Boot, Java, APIs
Work Culture	Collaborative, Innovative, Learning-Oriented
Focus Area	AI-powered solutions and automation tools
Role in Internship	Machine Learning Intern

4.3 Problem Statement

In today's competitive job market, candidates are required to apply to multiple job opportunities across various platforms, making the application process repetitive, time-consuming, and inefficient. Job seekers often need to manually search for relevant openings, customize resumes for different roles, and repeatedly fill out application forms on different websites. This not only increases effort but also reduces productivity and consistency in applications.

Additionally, many candidates lack access to personalized guidance for improving their resumes and preparing for interviews. Traditional systems do not provide real-time feedback on resume quality, skill gaps, or interview performance. As a result, candidates may apply for jobs without proper preparation, leading to lower success rates.

Another major challenge is the absence of an integrated platform that combines job application automation, career guidance, and interview preparation. Existing solutions either focus only on job listings or provide limited assistance in resume building and interview practice, without leveraging advanced AI capabilities.

Furthermore, reliance on external AI services introduces challenges such as API rate limits and system instability, which can disrupt user experience if not handled properly.

Therefore, there is a need for a comprehensive and intelligent system that can automate the job application process, provide personalized career recommendations, simulate interview environments, and ensure system reliability even under external limitations.

The proposed system, **"AI Auto-Apply Job Applicator & Career Coach"**, aims to address these challenges by integrating artificial intelligence with automation techniques to enhance efficiency, improve user experience, and increase the effectiveness of job applications.

4.4 Project Objectives

The primary objective of this project is to develop an intelligent and automated platform that simplifies the job application process while enhancing candidate preparation using Artificial Intelligence. The system aims to reduce manual effort, improve efficiency, and provide a comprehensive solution for job seekers by integrating automation, career guidance, and interview preparation into a single platform.

One of the key objectives is to **automate the job application process**, enabling users to apply to multiple relevant job opportunities without repetitive manual effort. The system is designed to identify suitable job roles based on the candidate's profile and streamline the application workflow.

Another important objective is to implement an **AI-based job matching system** that analyzes user skills and resume data to recommend appropriate job opportunities. This helps candidates focus on relevant positions and increases their chances of success.

The project also focuses on developing a **resume analysis module** that evaluates resumes and provides suggestions for improvement. This ensures better alignment with industry requirements and improves the likelihood of passing Applicant Tracking Systems (ATS).

In addition, the system aims to provide **personalized career guidance** by identifying skill gaps and recommending learning paths, enabling users to enhance their technical and professional capabilities.

The project further includes the development of a **voice-based mock interview system**, which simulates real interview scenarios and provides feedback on user responses, helping candidates improve their communication and confidence.

Another objective is to support **multiple resume management**, allowing users to maintain different resumes for various job roles and automatically select the most suitable one during applications.

Finally, the system emphasizes **reliability and user experience** by incorporating fallback mechanisms to handle AI API limitations and by providing an intuitive and interactive interface.

Overall, the project aims to create a smart, efficient, and reliable career assistance platform that enhances the job application process and supports users in achieving their career goals.

4.5 Scope of the Project

The scope of this project focuses on developing an intelligent and automated platform to assist job seekers in managing the end-to-end job application process efficiently. The system is designed to integrate multiple functionalities such as job application automation, resume analysis, career guidance, and interview preparation within a single unified platform.

The project primarily targets **students, fresh graduates, and job seekers** who face challenges in applying for jobs manually across various platforms. It enables users to upload their resumes, analyze their skills, and automatically apply to relevant job opportunities, thereby reducing time and effort.

The system also includes an **AI-based career guidance module**, which analyzes the user's profile and identifies skill gaps. Based on this analysis, it provides recommendations for improving technical skills and suggests relevant learning paths. This enhances the overall career readiness of the user.

Another important aspect within the scope is the **voice-based mock interview system**, which allows users to practice interview questions in a simulated environment. The system evaluates responses and provides feedback, helping users improve their confidence and communication skills.

The project supports **multi-resume management**, enabling users to maintain different resumes tailored to specific job roles or companies. This increases the effectiveness of applications by ensuring better alignment with job requirements.

From a technical perspective, the system is designed using a **scalable full-stack architecture**, with a Next.js frontend and a Spring Boot backend. It also integrates Artificial Intelligence services for automation and analysis. Additionally, fallback mechanisms are implemented to ensure system reliability even in cases of API limitations or failures.

However, the scope of the project is currently limited to a prototype-level implementation with simulated or limited real-time job data. Full integration with major job platforms such as LinkedIn or Indeed, as well as large-scale deployment, can be considered as future enhancements.

Overall, the project scope covers the development of a smart, efficient, and reliable career assistance platform that simplifies job applications and supports users in improving their employability.

4.6 Technologies and Tools Used

The development of the **AI Auto-Apply Job Applicator & Career Coach** system involves the use of modern technologies and tools across both frontend and backend, along with Artificial Intelligence and supporting libraries. These technologies ensure that the system is scalable, efficient, and user-friendly.

The **frontend** of the application is developed using **Next.js**, which provides a fast and optimized environment for building modern web applications. It supports server-side rendering and efficient routing, enhancing performance. The user interface is built using **React** with **TypeScript**, ensuring type safety and better code maintainability. For styling, **Tailwind CSS** is used to create a responsive and visually appealing design, while **Lucide React** provides lightweight and customizable icons.

The **backend** is developed using **Spring Boot**, a powerful Java-based framework for building RESTful web services. It handles business logic, API development, and server-side processing. **Spring Security** along with **JWT (JSON Web Token)** is used to implement secure authentication and authorization mechanisms, ensuring that only authorized users can access system features. For database interaction, **Spring Data JPA** is used, which simplifies data handling and object-relational mapping.

The system uses **H2 Database** during development for lightweight and fast testing, while it can be extended to **PostgreSQL** for production-level deployment. For resume processing, **Apache Tika** is utilized to extract text from PDF and DOCX files, enabling further analysis of user data.

Artificial Intelligence capabilities are integrated using the **Google Gemini API**, which is used for job matching, resume evaluation, and chatbot interactions. To ensure reliability, fallback mechanisms are implemented to handle API rate limits and failures.

Additionally, the system uses browser-based APIs such as the **Web Speech API**, which includes Speech Recognition and Speech Synthesis features for implementing the voice-based mock interview module.

Various development tools such as **Visual Studio Code** (for coding), **Git** and **GitHub** (for version control), and **Postman** (for API testing) are used during the development process.

1. Core Technologies (Development Stack)

These are the main building blocks of your system:

- Next.js
- React
- TypeScript
- Tailwind CSS
- Spring Boot
- Java
- Spring Security + JWT
- H2 / PostgreSQL

2. Supporting Technologies / Libraries

These enhance functionality:

- Google Gemini API (AI features)
- Apache Tika (resume parsing)
- Web Speech API (voice interview)
- Lucide React (icons)

3. Development Tools

These help you build and test:

- Visual Studio Code
- Git & GitHub
- Postman

4.7 System Architecture

The **AI Auto-Apply Job Applicator & Career Coach** system is designed using a modern, decoupled full-stack architecture that ensures scalability, modularity, and efficient data flow between components. The architecture follows a layered approach, separating the frontend, backend, and external services to maintain clear responsibilities and ease of maintenance.

At the **frontend layer**, the system is built using Next.js, which provides an interactive and responsive user interface. Users can access various features such as resume upload, job tracking, career guidance, and mock interviews through a centralized dashboard. The frontend communicates with the backend using RESTful APIs to send user requests and receive processed data.

The **backend layer** is implemented using Spring Boot and is responsible for handling business logic, API processing, and system workflows. The backend follows a structured design pattern consisting of Controllers, Services, and Repositories. Controllers handle incoming HTTP requests from the frontend, the Service layer processes business logic such as job matching, resume evaluation, and interview scoring, and the Repository layer interacts with the database to store and retrieve data.

The system integrates **Artificial Intelligence services** through a dedicated API service layer that communicates with the Google Gemini API. This layer is responsible for performing tasks such as job recommendation, resume analysis, chatbot responses, and interview evaluation. To ensure system reliability, fallback mechanisms are implemented to handle API rate limits or failures, allowing the system to continue functioning using alternative logic.

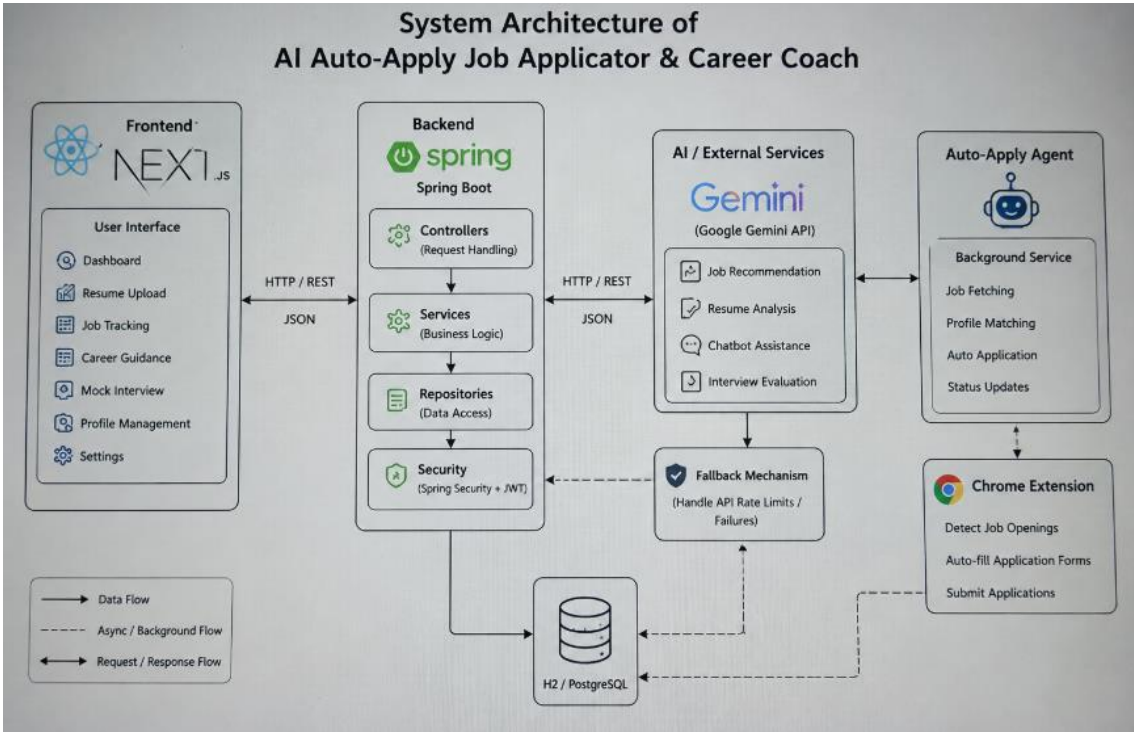
The **database layer** uses H2 Database for development and testing purposes, with support for PostgreSQL in production environments. It stores user profiles, resumes, job application data, and system logs required for tracking and analysis.

An important component of the architecture is the **Auto-Apply Agent**, which operates as an asynchronous background service. It continuously processes job listings, matches them with user profiles, and performs automated applications without blocking user interactions. This improves system performance and user experience.

Additionally, the system includes integration with a **Chrome Extension**, which interacts with external job portals to autofill application forms. This component enhances automation by enabling one-click job applications on supported platforms.

Overall, the system architecture ensures a clear separation of concerns, efficient communication between components, and reliable performance through the integration of AI services and asynchronous processing mechanisms.

Figure 4.2: System Architecture of AI Auto-Apply Job Applicator & Career Coach



4.8 Methodology

The development of the **AI Auto-Apply Job Applicator & Career Coach** system follows a structured and modular methodology to ensure efficient design, implementation, and testing of the application. The methodology is based on a systematic approach that includes requirement analysis, system design, development, integration, and evaluation.

The first phase involves **requirement analysis**, where the key challenges faced by job seekers are identified. This includes understanding the need for automating job applications, improving resume quality, and providing interview preparation tools. Based on these requirements, the overall objectives and features of the system are defined.

The next phase is **system design**, where the architecture of the application is planned. A layered architecture is adopted, separating the frontend, backend, and AI service layers. The data flow between components is defined, and suitable technologies such as Next.js, Spring Boot, and AI APIs are selected to ensure scalability and performance.

Following the design phase, the system moves to the **development phase**, where individual modules are implemented. The frontend is developed to provide an interactive user interface, while the backend handles business logic and API communication. Key modules such as resume upload and parsing, job matching, auto-apply agent, career guidance, and mock interview system are developed independently and then integrated.

The project also incorporates **AI integration methodology**, where external AI services (Gemini API) are used for tasks like resume evaluation, chatbot interaction, and job recommendations. To ensure reliability, fallback mechanisms are implemented to handle API failures and rate limits, allowing the system to continue functioning smoothly.

Another important aspect is the **asynchronous processing approach**, used in the auto-apply agent. Background services are utilized to perform job searching and application tasks without affecting the performance of the main system, ensuring a responsive user experience.

After development, the system undergoes **testing and validation**, where each module is tested individually and as part of the integrated system. Tools like Postman are used for API testing, and different scenarios are evaluated to ensure accuracy, performance, and reliability.

Finally, the system is refined through **evaluation and improvement**, where feedback is analyzed, and necessary enhancements are made to improve usability and efficiency.

4.9 Expected Outcomes

The **AI Auto-Apply Job Applicator & Career Coach** system is expected to significantly improve the efficiency and effectiveness of the job application process for users. By integrating automation and Artificial Intelligence, the system aims to reduce the time and effort required to search and apply for jobs manually.

One of the primary expected outcomes is the **automation of job applications**, where users can apply to multiple relevant job opportunities without repetitive manual input. This will help candidates save time and focus more on skill development and interview preparation.

Another key outcome is the **improvement in resume quality**. Through AI-based resume analysis, the system provides suggestions for optimizing content, improving keyword relevance, and aligning resumes with industry standards. This increases the chances of passing Applicant Tracking Systems (ATS) and getting shortlisted.

The system is also expected to deliver **personalized career guidance**, helping users identify their strengths and weaknesses. By analyzing skill gaps, the platform suggests appropriate learning paths, enabling users to enhance their technical and professional capabilities.

Additionally, the **voice-based mock interview module** is expected to improve users' confidence and communication skills. By simulating real interview scenarios and providing feedback, the system helps candidates prepare effectively for actual interviews.

Another important outcome is the **efficient management of multiple resumes**, allowing users to maintain different versions tailored to specific job roles. This ensures better alignment with job requirements and improves application success rates.

From a technical perspective, the system ensures **reliability and continuous performance** through the implementation of fallback mechanisms for handling AI API limitations. This guarantees that the application remains functional even under constrained conditions.

Overall, the project is expected to provide a **comprehensive, intelligent, and user-friendly platform** that simplifies job applications, enhances user preparedness, and increases the likelihood of successful job placement.

4.10 Certificates of completion



Cognifyz IT Solutions

Internship Completion Certificate

Date -06/07/2026

This is to certify that **Nishant Tomar (Intern ID: CTI/A1/C367943)** currently pursuing B.Tech from The IILM University, was working as a **Machine Learning Intern** with Cognifyz IT Solutions Pvt. Ltd. from 06/06/2026 to 06/07/2026.

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We extend our best wishes to Nishant Tomar for a successful future, and we have no doubt that he will continue to excel in the field of Machine Learning.


With Regards,
Cognifyz IT Solutions Pvt. Ltd.



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The following sources were referred to during the development of the project and preparation of this report:

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